Answer Key & Marking Scheme

Subject: Mathematics | **Test:** Test 2 (Day 14, 23 May 2026) — 2-Week Cumulative
Maximum Marks: 80

Syllabus: Chs 1–13: Real Numbers • Polynomials • Linear Equations in Two Variables • Quadratic Equations • AP • Triangles • Coordinate Geometry • Trigonometry • Applications of Trigonometry • Circles • Areas Related to Circles • Surface Areas and Volumes • Statistics

SECTION A — Answer Key (1 mark each)

Q1. (b) 1.

Q2. (b) 29 — $\alpha + \beta = -7$, $\alpha\beta = 10$. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 49 - 20 = 29$.

Q3. (c) infinitely many solutions — second equation is 3 × first.

Q4. (b) real and equal — $D = 16 - 16 = 0$.

Q5. (a) 154 — 11th term from end $= l - (n - 1)d = 184 - 10\cdot3 = 154$.

Q6. (b) 4/9 — ratio of areas = (ratio of sides)².

Q7. (a) $4\sqrt{2}$ — $\sqrt{(16 + 16)} = 4\sqrt{2}$.

Q8. (a) (–1, 7).

Q9. (a) 1 — $\frac{1}{2} + \frac{1}{2} = 1$.

Q10. (a) 3/4 — $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \tan^2\theta = 25/16 - 1 = 9/16$.

Q11. (c) $30/\sqrt{3}$ m $= 10\sqrt{3}$ m — $\tan 30° = h/30 \Rightarrow h = 30/\sqrt{3} = 10\sqrt{3}$.

Q12. (c) 2.

Q13. (c) 120° — sum of angles in the kite quadrilateral = 360°; opposite angles $90° + 90° + 60° + x = 360° \Rightarrow x = 120°$.

Q14. (a) 154 cm² — $\frac{1}{4} \times \pi \times 14^2 = 154$.

Q15. (a) $(4/3)r$ — $(4/3)\pi r^3 = \pi r^2\cdot h \Rightarrow h = (4/3)r$.

Q16. (a) 12.9 — $(2+3+5+7+11+13+17+19+23+29)/10 = 129/10 = 12.9$.

Q17. 0.4■ = 4/9.

Q18. $k = \pm6$ — $D = k^2 - 36 = 0$.

Q19. Mode = 3·Median – 2·Mean.

Q20. Circumference $= 2\pi r = 2 \times 22/7 \times 7 = 44$ cm.

SECTION B — Answer Key (2 marks each)

Q21. Adding: $348(x + y) = 1740 \Rightarrow x + y = 5$. Subtracting: $86(x - y) = 86 \Rightarrow x - y = 1$. Solving: **x = 3, y = 2**. (2)

Q22. a■ = a + 2d = 16 ; a■ – a■ = 2d = 12 ⇒ d = 6. Then a + 12 = 16 ⇒ a = 4. **AP: 4, 10, 16, 22, …** (2)

Q23. Section formula: $x = (3\cdot8 + 1\cdot4)/4 = 28/4 = 7$; $y = (3\cdot5 + 1\cdot(-3))/4 = 12/4 = 3$. Point: **(7, 3)**. (2)

Q24. LHS $= \sec^2\theta - \tan^2\theta = 1$ = RHS. ✓ (2)

Q25. Half-chord = 6. $r = 10 \Rightarrow d = \sqrt{(10^2 - 6^2)} = \sqrt{64} =$ **8 cm**. (2)

SECTION C — Answer Key (3 marks each)

Q26. *Proof.* Suppose $3 + 2\sqrt{5}$ is rational, say p/q ($q \neq 0$). Then $\sqrt{5} = (p/q - 3)/2 = (p - 3q)/(2q)$, which is rational (ratio of integers) — contradicting that $\sqrt{5}$ is irrational. Hence $3 + 2\sqrt{5}$ is irrational. (3)

Q27. Let zeroes be α and $1/\alpha$. Product $= \alpha\cdot(1/\alpha) = 1 = c/a = p/2 \Rightarrow$ **p = 2**. (3)

Q28. Equation: $y = 8 + 5(x - 1) = 5x + 3$ (for $x \geq 1$). For x = 16, y = 5·16 + 3 = ■83. (3)

Q29. a■ = S■ – S■■■ = $(3n^2 + 2n) - [3(n-1)^2 + 2(n-1)] = 3n^2 + 2n - (3n^2 - 6n + 3 + 2n - 2) = 6n - 1$. So a■■ = 6·25 – 1 = **149**. (3)

Q30. LHS $= (1 + \cot^2\theta)\cdot\sin^2\theta = \operatorname{cosec}^2\theta\cdot\sin^2\theta = (1/\sin^2\theta)\cdot\sin^2\theta = 1$ = RHS. ✓ (3)

Q31. Arc length L = r·θ (in radians). 60° = π/3 rad. So 20 = r·π/3 ⇒ r = 60/π ≈ **19.10 cm**. (3)

SECTION D — Answer Key (5 marks each)

Q32. Let height of building = h, distance between foot of tower and foot of building = d. From foot of tower: tan 30° = h/d ⇒ d = h√3. From foot of building: tan 60° = 50/d ⇒ d = 50/√3. Equating: h√3 = 50/√3 ⇒ h = 50/3 ≈ **16.67 m**. (5)

Q33. *Pythagoras Theorem:* Statement and proof. Consider right ΔABC right-angled at B. Draw BD ⊥ AC. Triangles ABD and ABC have ∠A common and ∠ADB = ∠ABC = 90°, so ΔABD ~ ΔABC ⇒ AB/AC = AD/AB ⇒ AB^2 = AC·AD. Similarly ΔBDC ~ ΔABC ⇒ BC^2 = AC·DC. Adding: $AB^2 + BC^2$ = AC(AD + DC) = AC·AC = AC^2. ✓ (5)

Q34. Volume of cone = (1/3)π·6^2·24 = 288π. Volume of hemisphere = (2/3)π·6^3 = 144π. Volume of remaining solid = 288π – 144π = 144π = 144 × 22/7 ≈ **452.57 cm³**. (5)

Q35. Compute means via class-marks. Mid-points x_i: 110, 130, 150, 170, 190. f_i: 12, 14, 8, 6, 10; Σf = 50.
Mean: Σf·x = 12·110 + 14·130 + 8·150 + 6·170 + 10·190 = 1320 + 1820 + 1200 + 1020 + 1900 = 7260. **Mean = 7260/50 = ■145.2.**
Median: cumulative frequencies 12, 26, 34, 40, 50. n/2 = 25 — falls in class 120–140. l = 120, cf = 12, f = 14, h = 20. Median = 120 + (25 – 12)/14 × 20 = 120 + 13×20/14 ≈ **■138.57**.
Modal class is 120–140 (highest frequency 14). Mode = l + (f■ – f■)/(2f■ – f■ – f■) × h = 120 + (14 – 12)/(28 – 12 – 8) × 20 = 120 + 2·20/8 = **■125**. (5)

SECTION E — Answer Key (4 marks each)

Q36. (i) AC: from (2,1) to (2,4), distance = √(0 + 9) = **3 m**. (1) (ii) BC: from (5,4) to (2,4), distance = √(9 + 0) = **3 m**. (1) (iii) Triangle has vertices (2,1), (5,4), (2,4). Area = ½|x_A(y_B – y_C) + x_B(y_C – y_A) + x_C(y_A – y_B)| = ½|2(4 – 4) + 5(4 – 1) + 2(1 – 4)| = ½|0 + 15 – 6| = ½·9 = **4.5 m²**. (2)

Q37. (i) Diagram: aeroplane at top P, two ships A and B, with depression angles 60° (to nearer ship A) and 30° (to farther ship B). Vertical from P meets ground at O. PO = 1500 m. (1) (ii) tan 60° = 1500/OA ⇒ OA = 1500/√3 = 500√3 m. tan 30° = 1500/OB ⇒ OB = 1500√3 m. Distance AB = OB – OA = 1500√3 – 500√3 = **1000√3** ≈ **1732 m**. (3)

Q38. (i) Cone height = total height – radius of hemisphere = 15.5 – 3.5 = **12 cm**. (1) (ii) Slant height ■ = √(r^2 + h^2) = √(3.5^2 + 12^2) = √(12.25 + 144) = √156.25 = **12.5 cm**. (1) (iii) TSA = curved surface of cone + curved surface of hemisphere = πr■ + 2πr^2 = π·3.5·12.5 + 2π·3.5^2 = π(43.75 + 24.5) = π·68.25 = 22/7 × 68.25 = **214.5 cm²**. (2)

— End of Answer Key —